Closest Pair of Points

Topic: DivideAndConquer File: d5.pdf

PROBLEM

Given points in a 2D plane, find the pair with the smallest Euclidean distance.

DIVIDE

Sort points by x-coordinate, split into left and right halves, and solve each half recursively.

COMBINE

Let delta be the smaller distance from the two halves. Consider points within delta of the dividing line, ordered by y-coordinate; only a bounded number of following points need comparison.

DISTANCE

For points (x1,y1) and (x2,y2), distance = sqrt((x1-x2)^2 + (y1-y2)^2). Comparing squared distances avoids square roots during comparisons.

COMPLEXITY

With careful sorting and merging, the algorithm runs in O(n log n) time and uses O(n) auxiliary space.

TAKEAWAY

The efficient combine step is what improves on checking all n(n-1)/2 pairs.